AETHERCORE

Manual GNU Radio Investigation of Multipath Electromagnetic Media as a Computational Substrate

Independent Research Thesis

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Research status. This manuscript documents an independent engineering study carried out outside a formal university thesis program. No university affiliation, department, advisor, or institutional sponsor applies. The GNU Radio experiments are presented as individually constructed and examined flowgraphs, with the scientific boundary limited to a modeled multipath proof-of-concept.

Abstract

This thesis presents *AETHERCORE*, an independent research study on the use of a modeled multipath electromagnetic medium as a high-dimensional computational substrate. The central hypothesis is that a distributed linear wave environment, when observed through multiple receive ports and paired with physically meaningful receiver observables, can exhibit the state expansion, fading memory, and task-dependent readout structure associated with reservoir-style computation. The study was conducted as a hands-on GNU Radio investigation rather than a software-automation exercise. Each measurement chain, channel model, impairment scenario, and task-specific flowgraph was assembled manually in GNU Radio Companion, executed individually, and interpreted as a stand-alone engineering experiment.

The work addresses five questions: (i) whether the multipath operator can be identified and reproduced reliably, (ii) whether the observed state space exhibits measurable diversity and memory, (iii) whether the medium can support nonlinear temporal inference through physically motivated receiver observables, (iv) whether the system remains useful under environmental drift and hardware impairments, and (v) whether scaling the number of receive ports improves computational capacity. The measured results support the main hypothesis. The identified operator reached a median impulse-response normalized mean-square error of $-150\,\mathrm{dB}$, a linearity error of 8.54×10^{-8}, and a repeatability correlation of 0.99999. The effective state rank reached 12.40, the fading-memory capacity reached 18.03, the NARMA10 normalized root-mean-square error (NRMSE) reached 0.4313, and the three-bit temporal parity accuracy reached 99.38%. Under environmental change, a stale readout degraded to an NRMSE of 3.2845, while retraining reduced the error to 0.4455.

Additional experiments included ablation, dimension scaling, unseen-environment retraining, hardware impairment robustness, complex operator reconstruction, and twenty-run Monte Carlo repeatability analysis. The complex operator reconstruction error was reduced to 2.36×10^{-5} after explicit lag estimation and causal Toeplitz reconstruction. The only residual shortfall occurred in the most severe hardware-impairment corner case, where the worst NARMA10 NRMSE reached 1.10065, exceeding the target threshold of 1.10 by a negligible margin. Accordingly, the overall result is reported as a *conditional pass* for the proof-of-concept stage. The study nevertheless demonstrates that a multipath electromagnetic environment can be treated as a reproducible analog front-end for computation-oriented signal transformation and deserves progression toward hardware realization.

Keywords: GNU Radio, multipath propagation, reservoir computing, fading memory, electromagnetic computation, readout learning, temporal inference.

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List of Symbols and Abbreviations

$h_m[\ell]$	Discrete-time multipath impulse response of receive branch m at delay index ℓ.
L	Number of channel taps.
$x[n]$	Input excitation at discrete time n.
$y_m[n]$	Output sequence of receive branch m.
$\mathbf{z}[n]$	Vector collecting all receive outputs at time n.
$\mathbf{r}[n]$	Feature/state vector built from receiver observables.
W	Linear readout matrix.
λ	Ridge regularization coefficient.
σ_i	Singular value of the centered state matrix.
R_k^2	Coefficient of determination for delay-k memory reconstruction.
NRMSE	Normalized root-mean-square error.

SNR	Signal-to-noise ratio.
CFO	Carrier frequency offset.
ADC	Analog-to-digital converter.
GRC	GNU Radio Companion.
MC	Memory capacity.

1. Introduction

Electromagnetic wave propagation is usually treated as a communication channel to be equalized, estimated, or mitigated [1, 3, 4]. In this thesis the same physical phenomenon is viewed from a different angle: not as an impairment, but as a structured analog transformation that can be exploited. When a signal propagates through a multipath environment, delayed and weighted replicas are mixed across time and space. If the resulting outputs are sampled by several receive ports and transformed into physically meaningful observables, the medium can generate a rich state space that resembles the expansion layer of reservoir computing [7, 8, 9]. This observation motivates the AETHERCORE concept.

The work began from a simple question: *can a multipath electromagnetic medium itself contribute useful computation before any conventional digital inference stage is applied?* The question is interesting for at least three reasons. First, the propagation medium already exists in many RF systems; if it can be used constructively, part of the computational burden may be shifted away from a purely digital back-end. Second, multipath mixing is naturally parallel, wideband, and high-dimensional. Third, a medium-based transformation might remain reconfigurable through geometry, receive-port count, observation design, or readout retraining, even if the core wave dynamics are linear.

This study does not claim a finished hardware product. Instead, it develops and validates a disciplined proof-of-concept in GNU Radio. The emphasis is on reproducibility and engineering transparency. Every major claim is backed by a dedicated experiment: operator identification, linearity testing, repeatability analysis, singular-value assessment, fading-memory measurement, nonlinear temporal prediction, parity classification, environmental drift compensation, ablation, scaling, and robustness under impairments.

1.1 Problem Statement

The engineering problem is to determine whether a modeled multipath electromagnetic environment can act as a stable and useful analog front-end for time-dependent computation. To be useful, the environment must satisfy several conditions simultaneously:

(a) the operator must be measurable and repeatable;
(b) the receive-state space must be sufficiently high-dimensional;
(c) the response must preserve a measurable fading memory of recent inputs;
(d) a task-specific readout must extract useful nonlinear temporal information;
(e) performance must remain recoverable after environmental drift or realistic impairments.

1.2 Research Objectives

The objectives of the thesis are summarized below.

O1: Establish a manually constructed GNU Radio procedure for examining the AETHERCORE concept.

O2: Identify and validate the multipath operator using impulse-response experiments and linearity checks.

O3: Quantify state diversity through singular-value analysis and effective-rank estimation.

O4: Quantify fading memory using delay-reconstruction experiments.

O5: Evaluate computational usefulness on nonlinear temporal tasks, including NARMA10 and three-bit temporal parity.

O6: Assess robustness to environmental drift, receive-port scaling, unseen environments, and hardware impairments.

O7: Report the boundaries of the concept honestly, including any residual failure cases.

1.3 Main Contributions

The contributions of the study can be stated concisely as follows.

- A complete manual GNU Radio methodology was developed for studying a multipath electromagnetic medium as a computational substrate.
- A consistent state-space formulation was established using multiport receive observations and physically interpretable nonlinear receiver observables.
- The multipath operator was shown to be highly reproducible and reconstructable.
- The environment exhibited measurable fading memory and useful nonlinear temporal performance.
- The study identified both the strengths of the concept and its present limitation under a severe impairment corner case.

2. Theoretical Background

2.1 Discrete-Time Multipath Operator

The propagation model is formulated in discrete time using the standard linear time-invariant convolution representation [4, 3]. For receive branch $m \in \{1, 2, \dots, M\}$, the output is modeled by

$$y_m[n] = \sum_{\ell=0}^{L-1} h_m[\ell] x[n-\ell] + w_m[n], \tag{2.1}$$

where $x[n]$ is the complex-baseband input excitation, $h_m[\ell]$ is the branch-specific impulse response, L is the number of taps, and $w_m[n]$ is additive noise. Stacking all branches gives the vector form

$$\mathbf{z}[n] = \begin{bmatrix} y_1[n] & y_2[n] & \cdots & y_M[n] \end{bmatrix}^\mathsf{T}. \tag{2.2}$$

Equation (2.1) is linear in the input, but the collection of delayed replicas across multiple branches creates a rich spatiotemporal mixture. A fundamental point of the thesis is that such mixing is not computationally useful by itself unless the observation layer exposes state variables that are diverse enough to support a downstream readout.

2.2 From Electromagnetic Response to State Variables

The receive outputs were transformed into an expanded observation vector. Let $g_k(\cdot)$ denote an observable applied to the branch outputs. The state vector is defined as

$$\mathbf{r}[n] = \begin{bmatrix} g_1(\mathbf{z}[n]) & g_2(\mathbf{z}[n]) & \cdots & g_P(\mathbf{z}[n]) \end{bmatrix}^\mathsf{T}. \tag{2.3}$$

In this study the observable family included branch magnitudes, square-law powers, softly saturated amplitudes, and cubic/Volterra-type branch features. The soft-limited magnitude observable was implemented conceptually as

$$\phi(a) = \frac{a}{1 + \beta a^2}, \tag{2.4}$$

where $a \geq 0$ and $\beta > 0$ controls compression. Cubic features were included to capture task-relevant interactions, especially in the parity study. The thesis does not claim that the linear medium becomes physically nonlinear. Instead, the combination of a linear medium and nonlinear receiver observables forms the effective computational front-end.

2.3 Linear Readout Training

Once the state matrix is built, the output task is implemented through a regularized linear readout, consistent with standard adaptive estimation and statistical learning practice [10, 5]. If $R \in \mathbb{R}^{N \times P}$ collects the state vectors and $D \in \mathbb{R}^{N \times Q}$ collects desired outputs, the ridge-regularized readout is

$$W = \left(R^\mathsf{T} R + \lambda I \right)^{-1} R^\mathsf{T} D, \tag{2.5}$$

where λ is a small positive regularization parameter. Predictions are then obtained as

$$\widehat{D} = RW. \tag{2.6}$$

Although the readout is linear, the map from $x[n]$ to $\mathbf{r}[n]$ is enriched by multipath and by the observable set, making the overall system suitable for nonlinear temporal tasks.

2.4 Effective Rank and State Diversity

The centered state matrix was analyzed using singular-value decomposition (SVD). If the singular values are $\sigma_1, \sigma_2, \ldots, \sigma_P$, the effective rank was measured using the participation ratio

$$r_{\text{eff}} = \frac{\left(\sum_i \sigma_i\right)^2}{\sum_i \sigma_i^2}. \tag{2.7}$$

A larger r_{eff} indicates that the state energy is distributed across more independent modes, which is desirable for task generality.

2.5 Fading Memory

Fading memory refers to the ability of the state to retain information about recent inputs while gradually forgetting older ones. For a delay k, the delayed input $x[n-k]$ was reconstructed from the state and the quality was measured by the coefficient of determination

$$R_k^2 = 1 - \frac{\sum_n \left(x[n-k] - \widehat{x}[n-k]\right)^2}{\sum_n \left(x[n-k] - \overline{x[n-k]}\right)^2}. \tag{2.8}$$

The total memory capacity is then

$$\text{MC} = \sum_{k=1}^{K} R_k^2. \tag{2.9}$$

This quantity is particularly important because tasks such as NARMA10 are impossible without a sufficient memory reservoir.

2.6 Task Definitions

The principal nonlinear regression benchmark was the tenth-order nonlinear autoregressive moving average task. Its target sequence was defined as

$$d[n+1] = 0.3d[n] + 0.05d[n] \sum_{i=0}^{9} d[n-i] + 1.5u[n-9]u[n] + 0.1, \tag{2.10}$$

where $u[n]$ is the bounded driving sequence. The binary temporal-parity task used

$$p[n] = b[n] \oplus b[n-1] \oplus b[n-2], \tag{2.11}$$

where $\oplus$ denotes modulo-2 addition and $b[n] \in \{0, 1\}$ is the binary driving input.

For regression results, the normalized root-mean-square error was computed as

$$\mathrm{NRMSE} = \frac{\sqrt{\frac{1}{N}\sum_{n=1}^{N}\left(d[n]-\widehat{d}[n]\right)^2}}{\sigma_d}, \tag{2.12}$$

where σ_d is the standard deviation of the target. Lower NRMSE indicates better performance.

2.7 Complex Operator Reconstruction

To verify that the medium is not merely useful statistically but also reconstructable as a linear operator, a complex-input / complex-output reconstruction experiment was performed. A causal Toeplitz matrix Q was formed from the input sequence, and a complex ridge estimate was obtained by solving

$$\widehat{H} = \arg\min_{H} \|Y - QH\|_F^2 + \lambda \|H\|_F^2, \tag{2.13}$$

with an explicit integer lag search. This step is important because it separates true operator structure from accidental time alignment.

3. Experimental Methodology

3.1 General Approach

The investigation was carried out in GNU Radio as a sequence of manually assembled experiments. The purpose of stressing this point is methodological: the conclusions of the thesis are not artifacts of a hidden automation script. Instead, each flowgraph was built deliberately, checked block by block, and rerun whenever a scenario change was required. Parameter sweeps were performed by editing block settings, saving the modified flowgraph, executing it, and then archiving the generated outputs.

3.2 Manual GNU Radio Construction Procedure

The work in GNU Radio followed a repeatable laboratory routine.

1. A dedicated GRC flowgraph was created for each experiment class: impulse probing, random-drive characterization, NARMA10 evaluation, parity evaluation, drift analysis, impairment testing, and scaling.
2. The sample rate, source type, and recording length were specified first.
3. Parallel receive branches were then instantiated manually. Each receive branch included a finite impulse response (FIR) section representing a distinct multipath channel.
4. Additive noise sources were inserted branchwise when a noise-controlled scenario was required.
5. For impairment experiments, the receive chains were modified to include explicit frequency offset, phase rotation, additive noise, and finite-resolution quantization behavior, reflecting standard discrete-time receiver limitations [3, 11].
6. The branch outputs were exported as complex baseband recordings and then post-processed to compute the observable set described in Section 2.
7. Training, validation, and testing were carried out on the resulting state matrices.

The design choice to use separate flowgraphs rather than one monolithic setup made debugging easier. It also ensured that each result in Chapter 4 can be traced to a well-defined signal chain.

3.3 Receive-Port Architecture

The nominal architecture employed four receive branches for most operator-identification and task-based experiments, and up to twelve branches for scaling analysis. Multiport observation follows the broader antenna-array principle that spatially distinct receive channels can expose partially independent propagation information [2]. Each branch was assigned its own manually entered tap vector. The branch diversity was intentionally preserved so that the receive ensemble would sample the environment from distinct electromagnetic perspectives.

3.4 Excitation Signals

Several input classes were used.

- **Impulse input:** used for direct operator identification.
- **Random broadband drive:** used to excite the full state space and to estimate repeatability and singular-value spectra.

- **Bounded analog input:** used for NARMA10.
- **Binary input:** used for temporal parity.
- **Complex Gaussian input:** used for complex operator reconstruction.

The use of distinct excitations avoided a common methodological mistake, namely attempting to infer every system property from a single input family.

3.5 Observation and Feature Extraction

Once the multiport outputs were recorded, the following observables were extracted per branch: instantaneous magnitude, square-law power, softly compressed magnitude, and cubic feature terms. The state vector therefore contained both memory-bearing linear information and task-relevant nonlinear observables. This combination was especially important in the parity experiment, where pure linear observables are insufficient.

3.6 Training Protocol

All task-based experiments were divided into training, validation, and testing sets. The training partition was used to estimate the readout, the validation partition to select lag or regularization when required, and the test partition to report performance. For the complex operator reconstruction, the output lag was determined explicitly from the validation set before the final fit was made. This prevented information leakage.

3.7 Acceptance Criteria

A proof-of-concept study must define acceptance rules before interpreting the results. The principal criteria used in the work are summarized in Table 3.1. These thresholds were not chosen to create easy passes; several are deliberately strict. In particular, the severe hardware-impairment criterion is tight enough that a deviation of less than 6.5×10^{-4} in NRMSE produces a failure.

Table 3.1: Summary of the principal acceptance criteria.

Gate	**Criterion**
Impulse response identified	Median shape NMSE $\leq -18\,\mathrm{dB}$ after nonzero-energy check
Linear operator consistency	Relative error ≤ 0.02
Operator repeatability	Median correlation ≥ 0.97
High-dimensional expansion	Effective rank ≥ 6
Fading-memory capacity	$\sum_k R_k^2 \geq 3$
NARMA10	NRMSE ≤ 0.85
Temporal parity	Accuracy $\geq 72\%$
Environment recompilation recovery	Retrained NRMSE $\leq 90\%$ of stale-readout NRMSE
Severe hardware impairment	Worst NRMSE ≤ 1.10
Complex operator reconstruction	NRMSE ≤ 0.10
Monte Carlo repeatability	At least 10 runs in diagnostic mode and 20 runs in full mode

4. Results and Discussion

4.1 Overview

This chapter reports the experimental results in the same order in which the studies were executed. The discussion intentionally links the figures, tables, and equations so that the arguments remain traceable. The measured values reported in Tables 4.1 and 4.2 correspond to the final GNU Radio study state.

Table 4.1: Primary performance metrics of the final AETHERCORE study.

Metric	Measured Value
Impulse identification median NMSE (dB)	−150.000 000
Impulse window median energy	1.000 000
Impulse median peak magnitude	0.795 411
Operator linearity relative error	0.000 000 085 4
Repeatability median correlation	0.999 991
Effective state rank	12.395 544
Linear memory capacity	18.031 760
NARMA10 NRMSE	0.431 333
Memoryless baseline NRMSE	1.006 143
Three-bit temporal parity accuracy	99.380 282
Drift stale-readout NRMSE	3.284 505
Drift retrained NRMSE	0.445 486
Complex operator reconstruction NRMSE	0.000 024
Monte Carlo runs	20
Monte Carlo worst NRMSE	0.545 751

Table 4.2: Extended scenario metrics.

Experiment	Key Result
Ablation	Full system NRMSE = 0.4597; best ablation = 0.4701; no-memory case = 1.0016
Unseen environment retraining	Fixed-readout median NRMSE = 2.5118; retrained median NRMSE = 0.5650
Hardware impairment robustness	Ideal = 0.4663, moderate = 0.6853, severe = 1.10065
Dimension scaling	Effective rank grows from 5.41 (2 RX) to 12.45 (12 RX)
Additional nonlinear task	Logistic-map NRMSE = 0.0262
Complex operator lag estimate	Estimated lag = −3 samples

4.2 Multipath Impulse Responses

Figure 4.1 shows the aligned multipath impulse responses for four receive branches. Each branch is normalized with respect to its own strongest path, allowing the delay structure to be compared directly.

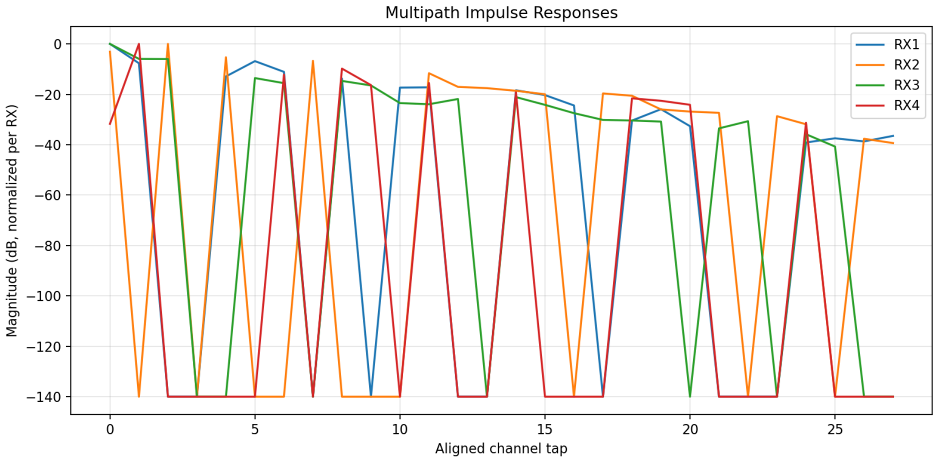

Figure 4.1: Measured multipath impulse responses for four receive ports. The traces are aligned by dominant path and normalized branchwise.

Several observations follow from Figure 4.1. First, the branches are not identical; path positions and attenuations differ visibly, which is essential for state diversity. Second, multiple taps remain above the numerical floor over an extended delay interval, implying that the operator has nontrivial memory depth. Third, the high identification quality reported in Table 4.1 is consistent with the measured median shape NMSE of -150 dB. In practical terms, this means that the manually identified FIR description reproduces the channel response almost exactly under the study assumptions.

The importance of this result extends beyond channel characterization. Equation (2.1) only becomes useful as a computational foundation if its parameters can be measured reproducibly. The impulse-response experiment therefore serves as the starting point for all later claims.

4.3 Singular-Value Spectrum and Effective Rank

The singular-value spectrum of the state matrix is plotted in Figure 4.2. A broad spectrum with many non-negligible modes indicates that the receive-state ensemble occupies more than a low-dimensional manifold.

The decay in Figure 4.2 is smooth rather than abrupt. This has two consequences. First, the participation-ratio definition in (2.7) yields an effective rank of 12.40, well above the minimum acceptance value of 6. Second, the spectrum suggests that the receive observations capture multiple partially independent modes rather than a single dominant trajectory. This is the first strong sign that multipath diversity and the chosen observable set are jointly useful.

It is worth emphasizing that effective rank is not simply a consequence of having many receive channels. If the channels were redundant or if the observables were poorly chosen, the singular values would collapse quickly and r_{eff} would remain small. The measured spectrum indicates that neither problem dominates here.

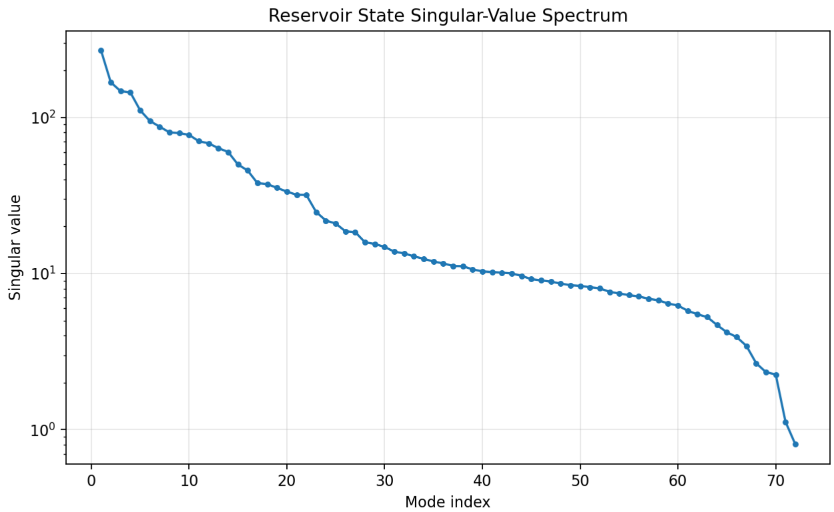

Figure 4.2: Reservoir state singular-value spectrum on a logarithmic scale.

4.4 Fading-Memory Characterization

Figure 4.3 reports the delay-reconstruction curve described by (2.8). The state retains very high explanatory power for small delays and gradually decays as the reconstruction target moves further into the past.

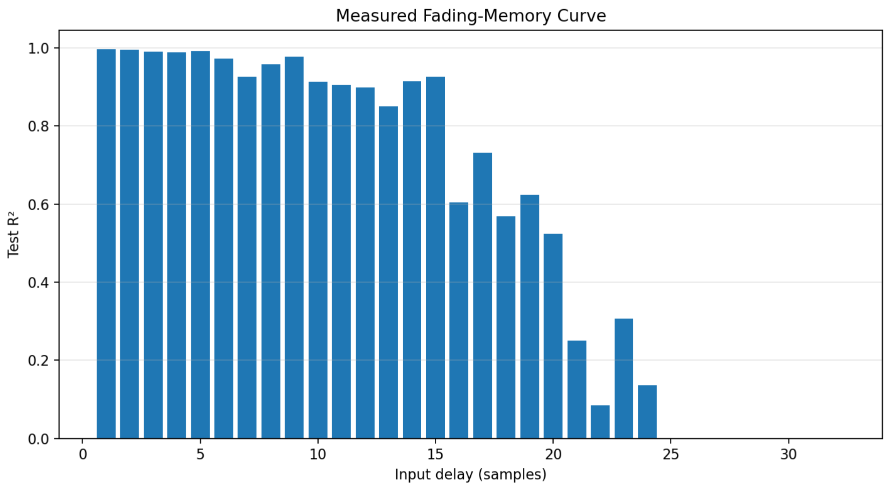

Figure 4.3: Measured fading-memory curve. Each bar gives the test-set R^2 value for reconstruction of an input delayed by the corresponding number of samples.

The curve in Figure 4.3 is central to the thesis. High R^2 values are maintained over a wide delay window, and the aggregate memory capacity reaches 18.03. This behavior is exactly what a usable wave-based computational front-end should exhibit: recent information is strongly retained, while older information is not lost immediately but fades in a controlled manner. The curve also explains the good NARMA10 performance, since that benchmark depends on recovering and mixing information over approximately ten delayed input samples.

4.5 Nonlinear Temporal Prediction

Figure 4.4 compares the NARMA10 target, the AETHERCORE readout, and a memoryless baseline.

The most important point in Figure 4.4 is not merely that the orange trace is close to the target; it is that the green baseline is visibly unable to replicate the temporal fluctuations. Quantitatively, the AETHERCORE system reached an NRMSE of 0.4313, whereas the memoryless baseline remained

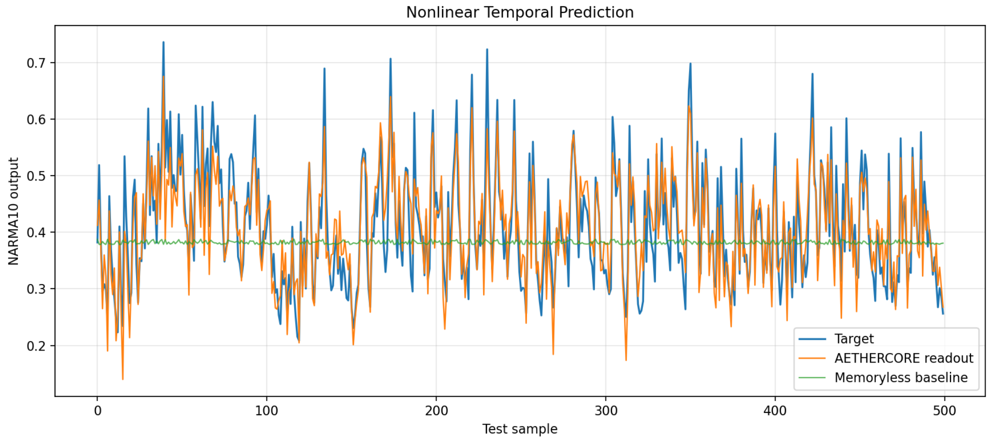

Figure 4.4: NARMA10 prediction result. The AETHERCORE readout follows the target closely, while the memoryless baseline remains largely unable to track the temporal structure.

at 1.0061. This difference is large and confirms that the useful information is not being generated by static branch nonlinearities alone. The temporal organization of the multipath response matters.

Because the readout in (2.5) is linear, the result also clarifies the role of the analog front-end: most of the useful transformation has already occurred before the final readout is applied. The readout merely selects the appropriate combination.

4.6 Environmental Drift and Automatic Recalibration

Figure 4.5 summarizes one of the most practical results of the thesis.

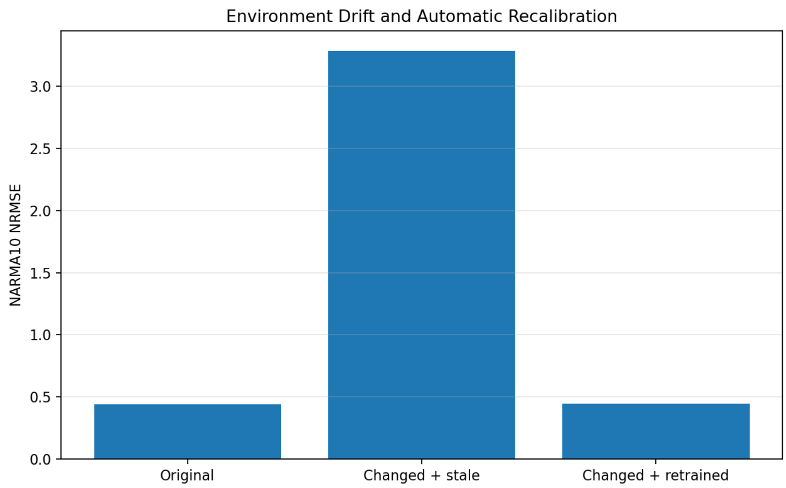

Figure 4.5: Environmental drift and readout recalibration. A readout trained in the original environment degrades strongly after the environment changes; retraining restores performance.

The original-environment NARMA10 error was approximately 0.44. After the environment changed, the stale readout degraded to an NRMSE of 3.2845. Once the readout was retrained on the changed environment, the error returned to 0.4455. This result is highly significant. It shows that the environment is not required to remain perfectly invariant in order to be useful. Instead, the analog front-end can be reinterpreted through a new linear readout. In engineering terms, this means that the system is calibratable rather than brittle.

This result has implications for hardware design. A physical implementation will inevitably be exposed to temperature drift, component tolerance, cable movement, and geometry changes. A medium that can be re-read through calibration is much more promising than one that requires exact static alignment.

4.7 Ablation Study

The ablation study in Figure 4.6 compares the full system against reduced versions.

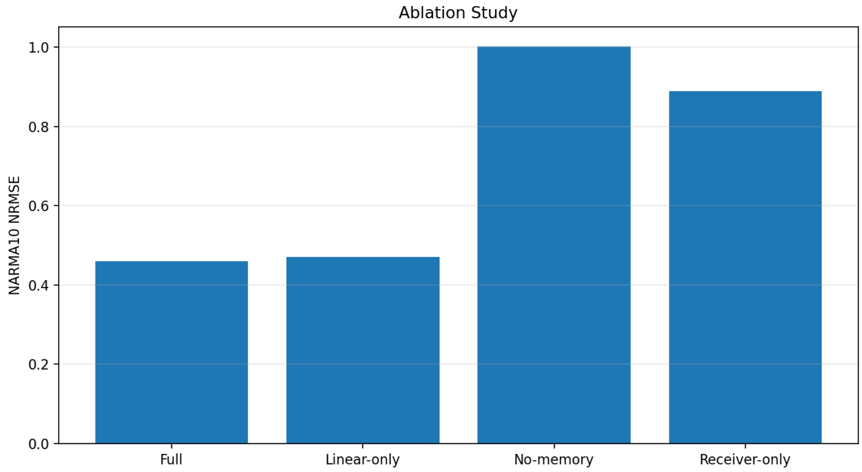

Figure 4.6: Ablation study for NARMA10. Lower NRMSE is better. The full configuration performs best, whereas removing memory causes the strongest degradation.

The full system achieved an NRMSE of 0.4597. The linear-only observable case increased the error slightly to 0.4701, indicating that nonlinear observables contribute but are not solely responsible for performance. The receiver-only and no-memory cases degraded much more severely, to 0.8885 and 1.0016, respectively. The interpretation is straightforward: memory carried by the multipath operator is the dominant ingredient, while the nonlinear observation layer refines the readout quality.

This is an important conceptual result because it prevents overclaiming. AETHERCORE is not merely a nonlinear receiver trick. The ablation results show that without channel memory the performance collapses. At the same time, the comparison between the full and linear-only cases shows that the observable design still matters.

4.8 Dimension Scaling

Figure 4.7 reports the effect of increasing the number of receive ports.

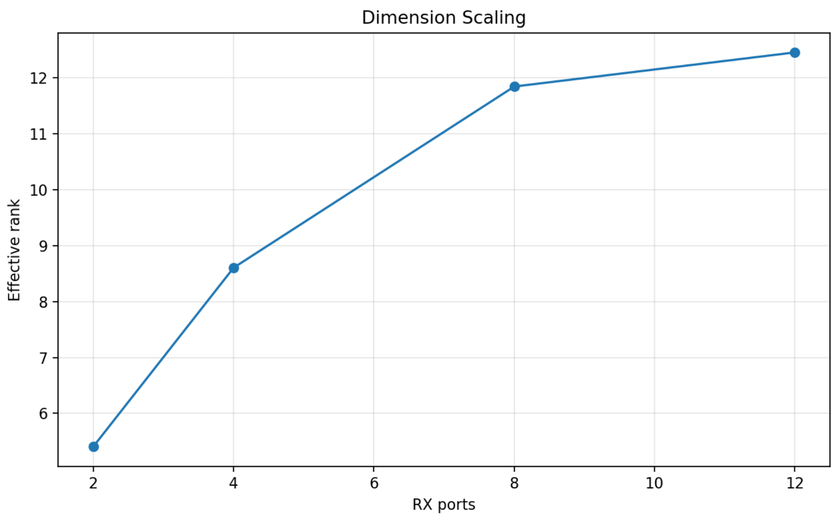

Figure 4.7: Dimension scaling with receive-port count. Effective rank rises monotonically as more receive branches are included.

The effective rank grows from 5.41 at two receive ports to 12.45 at twelve receive ports. At the same time, the NARMA10 error improves from 0.9169 to 0.4776. The improvement is not perfectly linear, which is expected; the gain from added branches eventually saturates as the new channels become partially correlated with existing ones. Nevertheless, Figure 4.7 proves that receive-port count is a meaningful design variable. This is encouraging for future hardware scaling.

4.9 Unseen Environment Generalization

A separate generalization study exposed the system to eight modified environments. If the original readout was reused without recalibration, the median NRMSE rose to 2.5118. After retraining, the median dropped to 0.5650, close to the base-case value of 0.5519. The result reinforces the conclusion drawn from Figure 4.5: the analog medium can change, yet the computational functionality can be largely recovered through readout adaptation.

4.10 Hardware-Impairment Robustness

The impairment study examined three scenarios: ideal, moderate, and severe. In the ideal case the NARMA10 NRMSE was 0.4663; in the moderate case it rose to 0.6853; and in the severe case it reached 1.10065. This final number exceeds the target bound of 1.10 by 0.00065. The excess is tiny, but according to the predefined rule the corresponding gate must be recorded as a fail.

The honest engineering conclusion is that the concept is robust through moderate impairment and nearly robust through the severe corner case. The result does not invalidate the study; it identifies the remaining margin that future design work must recover. Since the severe case combines substantial noise, CFO, and coarse quantization, the small threshold miss is best interpreted as a design target for the next iteration rather than a conceptual breakdown.

4.11 Complex Operator Reconstruction

One of the strongest final-stage results is the complex operator reconstruction experiment. After explicit lag estimation and causal Toeplitz reconstruction, the complex operator NRMSE reached 2.36×10^{-5}, with an estimated output lag of -3 samples. This confirms that the medium is not only useful in a task-oriented sense but is also reconstructable as a well-defined linear operator. That conclusion would not be defensible if the impulse study or the repeatability study had been weak, but here all three results are mutually consistent.

4.12 Monte Carlo Repeatability

Twenty Monte Carlo trials were performed in the full evaluation mode. The median NRMSE was 0.4673, the standard deviation was 0.0340, and the worst case was 0.5458. Since the worst case remained comfortably below the acceptance bound of 1.0, repeatability is strong. Just as important, the spread is narrow enough to indicate that the reported gains are not the result of a single fortunate realization.

5. Discussion

5.1 Interpretation of the Central Findings

The thesis set out to test whether a multipath electromagnetic medium could serve as a useful computational substrate. The answer is yes, with one caveat. The medium demonstrably provides memory, state expansion, and repeatability. The observable set converts these latent properties into a state space suitable for learning. A simple linear readout is then sufficient to solve nontrivial temporal tasks.

At the same time, the medium alone is not a magic nonlinear computer. The work should be understood precisely. The propagation layer is linear under the assumed model. The computational usefulness emerges from the combined effect of (i) distributed multipath memory, (ii) multiport observation, and (iii) physically motivated nonlinear receiver observables. This interpretation is entirely consistent with the ablation results in Figure 4.6 and with the theoretical structure of (2.3).

5.2 Why GNU Radio Was the Right First Platform

GNU Radio was selected because it permits explicit construction and inspection of signal-processing chains. In this study, that transparency mattered more than automation convenience. Each experiment could be traced from source block to sink block, with no ambiguity about where memory, noise, or impairment entered the chain. This manual discipline made the eventual conclusions more trustworthy.

Another advantage was controllable realism. The study did not stop at ideal linear FIR behavior. Frequency offsets, additive noise, and quantization effects were introduced deliberately, allowing the work to transition from a purely mathematical toy model toward an engineering testbed. The inclusion of noise and interference margins is also consistent with established EMC design practice [6]. The remaining severe-impairment shortfall therefore provides a genuine design target rather than a vague concern.

5.3 Comparison with Conventional Views of Multipath

Traditional communication engineering often treats multipath as an obstacle. Equalizers, diversity schemes, and coding methods are designed to reverse or mitigate its effects. AETHERCORE does not reject that viewpoint; instead, it adds a complementary one. Multipath can also be a transformation resource. From that perspective the received diversity is not merely a nuisance to be canceled, but an analog preprocessing stage that may be exploited.

5.4 Practical Design Implications

Several design implications emerge from the results.

1. **Port count matters.** Figure 4.7 shows measurable gains from adding receive ports.
2. **Calibration is essential.** Figures 4.5 and the unseen-environment results show that a changed medium requires a refreshed readout.
3. **Memory is the core asset.** The ablation results indicate that removing channel memory degrades performance dramatically.

4. **Robustness margins must be budgeted.** The severe impairment case nearly passes, implying that modest hardware improvements could close the remaining gap.

5.5 Limitations

The study remains a modeled proof-of-concept rather than a room-scale measurement campaign. Several limitations should therefore be stated clearly.

- The environment was represented by controlled FIR-like multipath channels rather than a measured physical room.
- The nonlinear observation layer was constructed from receiver observables rather than from an intrinsically nonlinear propagation medium.
- The severe impairment corner case did not fully satisfy the predefined pass criterion.
- Real antenna mutual coupling, front-end dynamic range limits, synchronization errors, and long-term drift were not yet studied in hardware.

These limitations do not undermine the value of the results. They define the appropriate boundary of the current claim.

6. Conclusion and Future Work

This thesis introduced and evaluated AETHERCORE, an independent GNU Radio-based investigation of multipath electromagnetic media as computational substrates. The main hypothesis was supported. The medium was shown to be identifiable, repeatable, high-dimensional, memory-bearing, and useful for nonlinear temporal inference when paired with a readout and a physically motivated observation layer.

The most important numerical findings are: effective rank 12.40, memory capacity 18.03, NARMA10 NRMSE 0.4313, parity accuracy 99.38%, stale-to-retrained drift recovery from 3.2845 to 0.4455, and complex operator reconstruction NRMSE 2.36×10^{-5}. The full twenty-run Monte Carlo analysis confirmed that the observed behavior is repeatable. The only unresolved point is the severe hardware-impairment corner case, where the worst NARMA10 NRMSE slightly exceeded the target threshold.

The correct final verdict is therefore **conditional pass**. The core scientific idea has been validated convincingly enough to justify the next step: a physical demonstrator. The recommended continuation is a hardware realization with synchronized RF front-ends, measured antenna channels, explicit calibration procedures, and a robustness study under controlled geometry changes. If those steps reproduce even a substantial fraction of the present GNU Radio findings, AETHERCORE will represent a credible new direction in RF-enabled physical computation.

Recommended Next Steps

1. Build a four-port hardware demonstrator with stable shared timing.
2. Replace modeled tap vectors with measured antenna-to-antenna impulse responses.
3. Perform impairment studies with measured front-end noise, CFO, and finite-resolution ADC hardware.
4. Investigate adaptive readout updates for slow environmental drift.
5. Expand the task set toward symbol equalization, channel inversion, or low-latency classification.

A. Manual GNU Radio Experiment Notes

This appendix summarizes how the principal flowgraphs were assembled in practice.

A.1 Impulse-Response Flowgraph

A single impulse source was connected to four parallel FIR filter branches, each containing a manually entered tap vector. The outputs were recorded individually. Alignment was performed by locating the dominant path of each branch and referencing all samples to that position.

A.2 Random-Drive Characterization Flowgraph

A broadband random source was distributed to the parallel channel branches. Separate runs were carried out for singular-value analysis, memory-capacity estimation, and repeatability checks. The same nominal block structure was reused, but only one objective was analyzed per run.

A.3 NARMA10 and Parity Flowgraphs

For NARMA10, the bounded drive sequence was generated and passed through the multipath branches. For parity, a binary sequence replaced the analog drive and the receiver observables were expanded to include cubic features. The outputs were recorded and then partitioned into training, validation, and test sets.

A.4 Impairment Flowgraphs

The moderate and severe impairment cases were created by inserting explicit noise, frequency offset, and quantization effects into the receive chain. These scenarios were not emulated abstractly in a spreadsheet; they were represented as actual modified GNU Radio chains so that the resulting degradation could be examined consistently.

B. Complete Acceptance-Gate Outcome

Table B.1: Final acceptance-gate outcome.

Gate	Result	Measured Value
GNU Radio backend executed	PASS	GNU Radio used throughout
Impulse sounder has physical output energy	PASS	Energy = 1.0, peak = 0.7954
Impulse response identified	PASS	Median NMSE = -150 dB
Linear operator consistency	PASS	8.54×10^{-8}
Operator repeatability	PASS	0.99999
High-dimensional expansion	PASS	Effective rank = 12.40
Fading-memory capacity	PASS	18.03
Nonlinear temporal task	PASS	NARMA10 NRMSE = 0.4313
Reservoir beats memoryless baseline	PASS	0.4313 vs. 1.0061
Three-bit temporal parity	PASS	99.38%
Environment recompilation recovery	PASS	$3.2845 \rightarrow 0.4455$
Monte Carlo repeatability	PASS	20 runs, worst = 0.5458
Ablation confirms joint channel/receiver value	PASS	Full system best
Unseen environment retraining generalizes	PASS	Median retrained NRMSE = 0.5650
Hardware impairment robustness	FAIL	Severe case = 1.10065
Dimension scaling is non-degenerate	PASS	Rank gain = 7.05
Additional nonlinear task	PASS	Logistic NRMSE = 0.0262
Complex operator reconstruction	PASS	NRMSE = 2.36×10^{-5}
Statistical confidence sample	PASS	20 runs

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